

INDUSTRIAL INTERNSHIP REPORT

Title: Implementation of a Microcontroller on an FPGA Board

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INTRODUCTION:

Microcontrollers form the backbone of modern embedded systems and are widely used in industrial automation, consumer electronics, and communication systems. Understanding their internal architecture is essential for engineers working in embedded systems and VLSI design.

This internship project focuses on the implementation of a basic microcontroller on an FPGA board using Verilog HDL. The work aims to bridge the gap between theoretical processor architecture concepts and real hardware implementation. The project was developed and simulated using Xilinx ISE, targeting an FPGA development platform.

Objective:

The main objectives of this project are:

- * To study the internal working of a microcontroller
- * To design a basic 8/16-bit microcontroller architecture
- * To implement modular hardware design using Verilog HDL
- * To design an FSM-based control unit
- * To verify functionality through simulation and waveform analysis
- * To understand FPGA-based design methodology

SYSTEM ARCHITECTURE:

The implemented microcontroller follows a multi-cycle architecture, where each instruction is executed in multiple clock cycles. The major blocks of the system are:

- * Program Counter (PC)
- * Instruction Memory
- * Instruction Register (IR)

- * Instruction Decoder
- * Control Unit (FSM)
- * Register File
- * Arithmetic Logic Unit (ALU)
- * Data Memory
- * GPIO Interface

All modules were designed independently and integrated into a top-level microcontroller module.

MODULE IMPLEMENTATION:

A. Program Counter

The Program Counter is an 8-bit register that holds the address of the current instruction. It increments during each instruction fetch cycle and supports reset and enable control.

B. Instruction Memory

A ROM-based instruction memory of size 256×16 -bit was implemented. Instructions are fetched using the PC address and supplied to the instruction register.

C. Instruction Register

The instruction register stores the fetched instruction and is enabled by the `ir_en` signal from the control unit.

D. Instruction Decoder

The decoder extracts the opcode, source registers, destination register, and immediate values from the instruction word. These decoded signals guide the datapath and control logic.

E. Control Unit

The control unit is designed as a Finite State Machine with the following states:

- * Fetch

- * Decode

- * Execute

- * Write Back

It generates control signals such as pc_en, imem_en, ir_en, reg_wr, and alu_op.

F. Register File

A 16×16 -bit register file was implemented with two read ports and one write port. Register writing occurs during the write-back stage.

G. Arithmetic Logic Unit

The ALU performs arithmetic and logical operations such as addition, subtraction, and logic functions based on the opcode.

H. Data Memory

A simple RAM module was implemented to support load and store operations.

I. GPIO Interface

A basic GPIO module was integrated to demonstrate interaction with external peripherals.

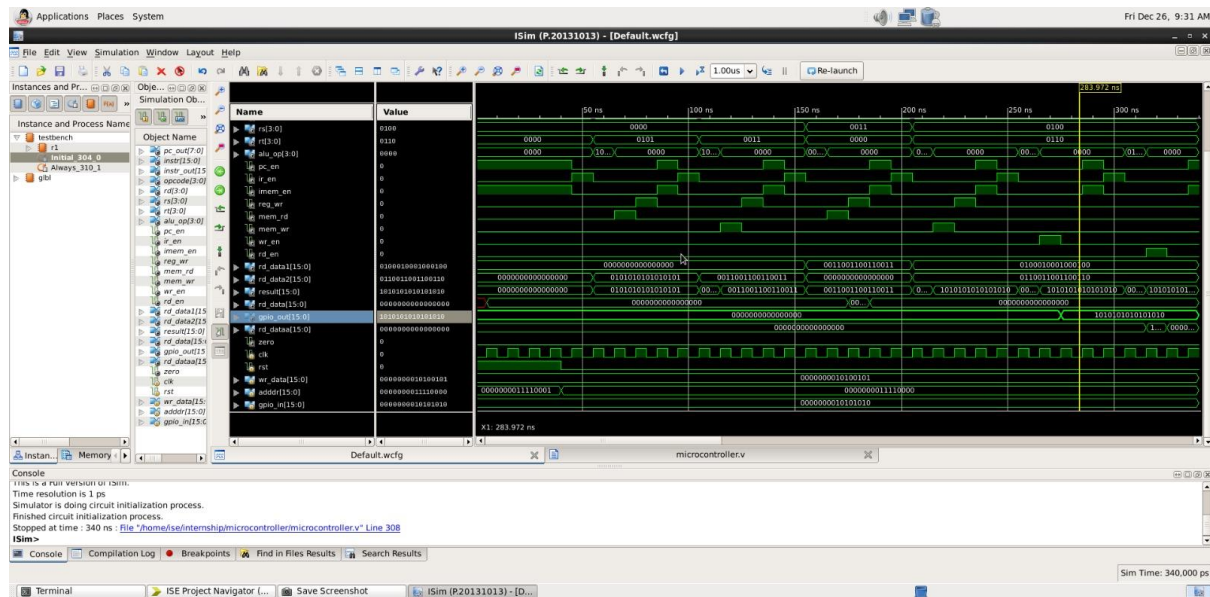


Fig. 7. Simulation waveform

RESULTS AND DISCUSSION:

The microcontroller design was successfully simulated, and all modules functioned as expected. Instruction execution followed correct sequencing under FSM control. The modular design approach simplified debugging and integration.

LEARNING OUTCOMES :

This project enhanced understanding of:

- * Microcontroller architecture
- * Verilog HDL coding
- * FSM-based control design
- * Datapath and control separation
- * FPGA-based processor implementation
- * Simulation and waveform debugging

CONCLUSION:

The implementation of a microcontroller on an FPGA board provided valuable insight into processor design at the hardware level. This project strengthened practical knowledge of digital system design and FPGA-based implementation, laying a strong foundation for advanced embedded systems and VLSI applications.

FUTURE SCOPE:

- * Integration of UART for serial communication
- * Support for interrupts and branching instructions
- * Implementation of pipelined architecture
- * Deployment on physical FPGA hardware with real I/O devices

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Reference:

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